

given that $\partial\pi_i/\partial b_i > 0$ (see Lemma A.1), π_i decreases for this agent. As a consequence $U_i(\mathcal{D})$ increases but $U_i(\mathcal{I})$ remains the same, so agent i (strictly) prefers to dislike. Similarly, because $\partial\pi_i/\partial p < 0$ and $\partial\pi_i/\partial q < 0$ for $b_i < 1/2$ (see Proposition 3), π_i decreases for this agent (making $a_i = \mathcal{D}$ a best response). In both cases, we see that r_P increases. ■

A.5 Proofs from Section 6

Proof of Proposition 5. For part (a), let $V > 0$ be the virality of any given article before the provenance policy. Note by Lemma A.4 that $\kappa \leq c/N$, so in particular it is never a best response for agent i to play $a_i = \mathcal{S}$ when $\pi_i = 0$. Conditional on an article containing misinformation and given provenance policy ρ , the seed agent's probability of sharing it is bounded above $1 - \rho$, and the virality of the misinformation article is thus bounded above by $(1 - \rho)N$. Setting $\rho^* = (N - V)/N$ guarantees that the provenance policy reduces the virality of misinformation.

For part (b), we expand on some of the details in Example 1, where we fix an article that necessarily contains misinformation. All user beliefs are uniformly distributed on $[0, 2/3]$, the message is $m = R$, $\phi(r) = 1/2$, $p = 0.7$, and $q = 0.3$. With no provenance policy, we note that users with beliefs $b_i > 1/2$ share the article and those with beliefs $b_i < 1/2$ do not. Each user i sharing the article with probability $\zeta = 1/4$, the total virality of the article can thus be computed simply as $\sum_{i=1}^N \zeta^i \approx \zeta/(1 - \zeta) = 1/3$ for large N .

Next, we consider a provenance policy with $\rho = 1/4$, so with probability $1/4$ each user (independently) will fact-check the content. Given our previous assumption, if a user fact-checks she will share when the fact-check comes back as $\nu = \mathcal{T}$ and will not share when the fact-check comes back $\nu = \mathcal{M}$. We suppose the article starts at seed agent 1.²⁶ User 1, similar to the no provenance policy, will share the article with probability $\zeta = 1/4$ if there is no fact-check. On the other hand, user 1 will share the article with certainty if there is a fact-check and it comes back $\nu = \mathcal{T}$, but will ignore the article with certainty if there is a fact-check and it comes back $\nu = \mathcal{M}$. Thus, user 1 shares the article with probability $\zeta(1 - \rho) = 3/16$. Any user $i \geq 2$ will know the article might have been previously fact-checked, and therefore will assign higher likelihood to its veracity than π_i defined in equation (2). In particular, user 2 will assign likelihood $\pi'_2 = \frac{(\rho + (1 - \rho)\zeta)\pi_2}{\rho\pi_2 + (1 - \rho)\zeta} > \pi_2$ that the article is truthful. For our parameter values, one can verify that $\pi'_2 > 1/2$ regardless of $b_i \in [0, 2/3]$ and so user 2 shares the article with probability 1 if she does not fact-check. Thus, the probability that agent 2 (and agents thereafter) shares is $1 - \rho = 3/4$. This implies that virality is $\sum_{i=2}^N (1 - \rho)^{i-1} \approx (1 - \rho)/\rho = 3$ for large N . Finally, the virality of the article after the provenance policy is $3/16 + 3/16 \cdot 3 = 3/4 > 1/3$. Thus, the provenance policy increases the virality of a misinformation article.

For part (c), we note that the generous bound in part (a) still applies identically even if the platform chooses a different sharing network following the provenance policy. ■

²⁶We do not claim that this is an optimal seeding strategy for the platform, but establishes a lower bound on the engagement the platform can obtain, which is sufficient to prove that the provenance policy can backfire.